LESSON 7.8
SIMPLE INTEREST

LESSON INTRODUCTION

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multistep real-world percentage problems.

LEARNING TARGETS

- I can solve problems that involve simple interest.
- I can compare simple interest and compound interest.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How do you calculate simple interest?
- Is the interest accrued on loans or investments always simple interest?

LESSON NARRATIVE

In this lesson, students are introduced to simple interest, including how to calculate it to determine the interest accrued on investments or loans using the simple interest formula. They then use this to solve real-world problems involving simple interest. The lesson also includes a component where students compare simple and compound interest. The focus is not on calculating or fully understanding compound interest but rather recognizing that simple interest is not the only type of interest used in the real world.

COMPONENT SUMMARY

7.8.1 WARM-UP *(~5 MINS.)*

Identify and correct an error on a receipt

7.8.2 GUIDED INSTRUCTION *(15 MINS.)*

Develop an understanding of simple interest

7.8.3 TRY IT *(5 MINS.)*

Practice calculating simple interest

7.8.4 COLLABORATION *(15 MINS.)*

Compare simple and compound interest

7.8.5 WRAP-UP *(~5 MINS.)*

Demonstrate mastery of learning target

RESOURCES

GLOSSARY

Key Terms:

- Simple interest

Additional Terms:

- Compound interest

MATERIALS

- (Optional) Chart paper & markers

ADDITIONAL PREPARATION

- Consider having a guest speaker who works in finance join your class virtually or in person to discuss simple and compound interest applications.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to recognize when to find the interest only versus when to add the accrued interest to the principal when solving problems.

THINGS TO CONSIDER

- Allowing students to use calculators during the lesson will help students focus on discovery and process of calculating simple interest and using it to solve problems. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 7.8.4 Collaboration, students discuss the similarities and differences in simple and compound interest. Ask students to initially complete the problems independently and then discuss their responses with their partners. These discussions will build student confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Real-World Contexts

7.8.3 Try It provides students with opportunities to solve real-world simple interest problems. During the debrief, ask a variety of students to share their work. Engage students in a discussion about the utility of the mathematics to ignite student interest.

DIFFERENTIATE INSTRUCTION

7.8.4 Collaboration provides both auditory and visual entry points into the activity. Encourage students to use the tables as visual supports and to actively discuss the strategies they use.

For the 7.8.5 Wrap-Up, allow students to work a simple interest example or use speech-to-text software to demonstrate their mastery of the lesson learning targets.

7.8.1 WARM-UP: MATH FAIL

QUESTIONING

Consider asking students to bring in a receipt from a store or restaurant and allow them to show the class how the tax and/or tip was calculated.

7.8.1 Warm-Up: Math Fail

The cost for dinner at a restaurant was $23.60. As a courtesy, the restaurant listed suggested tips for customers on the receipt. The receipt stated a 15% tip was $6.53, an 18% tip was $7.83, and a 20% tip was $8.70. The customer underlined the recommended 15% tip amount and wrote, "No it's not."

1. Explain whether the restaurant or customer is correct. Justify your answer with calculations.
   The customer is correct because $23.60 times 0.15 is $3.54 which they rounded to $3.50.

7.8.2 GUIDED INSTRUCTION: SIMPLE INTEREST

7.8.2 Guided Instruction: Simple Interest

1. Kristina received $100 in a birthday card and wants to put the money into her savings account, where she will earn an additional 2% each year it remains in her account. This is known as ***simple interest***.

> ***Simple interest*** is a method of computing interest. Interest is computed from the (original) principal alone, no matter how much money has accrued so far.

   a. How much is 2% of $100?
   $2.00

   b. How much money will be in Kristina's savings account after one year if she makes no additional deposits or withdrawals after putting in her birthday money?
   $102.00

   c. Compare your answer with your partner and discuss how you determined the answer for Part B.
   I knew that Kristina had $100 already and then earned an additional $2 which equals $102 when added together.

2. Suppose an investment of $100 earns $3.50 interest in one year.

   a. What investment amount will earn $61.60 in two years at the same rate?
   $0.035x = \frac{61.60}{2}$; $x = \$880.00$
   Students could also set up a percent proportion.

   b. Ask two other partner pairs to describe how they found the initial investment. Record their initial investment and write your summary of their process. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Initial Investment	My Summary of Their Reason	Initials
$880.00	*Answers will vary. They divided $61.60 by 2 since there was 2 years and then found what amount $30.80 is 3.5% of.*	
$880.00	*Answers will vary. They found what amount 3.5% is of $61.60 and then divided by 2.*	

   c. Review the initial investments and reasons provided by other partner pairs. If necessary, fix any mistakes you and your partner made.

> **Simple Interest Formula**
>
> $$I = p \times r \times t$$
>
> where I is the interest earned, p is the principal starting amount of money, r is the interest rate per year, and t is the length of the investment.

3. Tonio took out a six-year car loan for $15,600, with a 3% simple interest rate over the life of the loan.

   a. How much interest, in dollars, will Tonio pay annually?
   $I = (15{,}600)(0.03)(1) = \$468.00$ *per year*

   b. How much will Tonio pay in interest over the course of **six** years?
   $I = (15{,}600)(0.03)(6) = \$2{,}808.00$

TEACHER MOVES

Ask students what they know about interest. List examples on the board. Explain that financial institutions charge interest for allowing a customer to borrow money with a loan and pays a customer interest for keeping their money in a savings account. Answer any questions students have about the concept of interest.

ENRICHMENT

Consider asking students the following:

- How did you determine the initial investment amount for the previous question?
- How does the method you used relate to the simple interest formula?
- How could you use the formula to determine an interest rate or the length of the investment?

QUESTIONING

Consider allowing students to "shop" for a car locally and use the advertised price and a current, realistic percentage rate to see how much interest they would have to pay over the life of a 5-year loan.

7.8.2 GUIDED INSTRUCTION: SIMPLE INTEREST (CONTINUED)

c. What is the total amount Tonio will pay back at the end of his six-year loan?
$15{,}600 + 2{,}808 = \$18{,}408.00$

d. Explain why the total amount Tonio will have paid back by the end of his loan is more than the loan amount he borrowed.
He will pay back more because he has to pay back the amount he borrowed, plus the interest accrued on the loan.

7.8.3 TRY IT: SIMPLE INTEREST

ENRICHMENT

- What components of the simple interest formula are given in the question 1? Question 2?
- What is unknown?

TEACHER MOVES

Consider suggesting students use estimation to check the reasonableness of their answers:

- Estimate the interest using mental math before using the simple interest formula to calculate the precise amount. How close is your answer to your estimate?

DIFFERENTIATE INSTRUCTION

Consider allowing students to work with a partner for additional support and an auditory entry point. Display the simple interest formula with examples to provide a visual entry point.

7.8.3 Try It: Simple Interest

1. Quana is researching rates for investing $\$3{,}500$ over a 3-year period.

   How much money would Quana earn on an investment that offers a 4% simple interest rate?
   $\$420.00$

2. Sanjay has $\$750$ he wants to put into a savings account.

   How much would he have at the end of 5 years in an account that offers a 2% simple interest rate?
   $\$825.00$

7.8.4 Collaboration: Comparing Interest

Not all interest is simple interest; there is also ***compound interest***. Compound interest is interest earned on interest over time.

With your partner, complete the following.

1. Charlie wants to take out a $5,750 loan for new furniture. The bank offers him two different options for his loan. He can take out a loan for three years at a rate of 4% compounded annually, or he can take out a loan for four years at 5% simple interest.

   When a loan is compounded, that means interest owed each year is based on the principal *and* any accrued interest, not the principal value alone.

   a. The compound interest table has been filled out. Complete the simple interest table to help Charlie choose the loan he will pay the least.

4% Compound Interest	
Year 1	$5750 \cdot 1.04 = 5{,}980$
Year 2	$5750 \cdot 1.04 \cdot 1.04 = 6219.20$
Year 3	$5750 \cdot 1.04 \cdot 1.04 \cdot 1.04 = 6{,}467.97$

5% Simple Interest	
Year 1	$5750 \cdot 0.05 \cdot 1 = 287.50$ $287.50 + 5750 = \$6{,}037.50$
Year 2	$5750 \cdot 0.05 \cdot 2 = 575$ $575 + 5750 = \$6{,}325$
Year 3	$5750 \cdot 0.05 \cdot 3 = 862.50$ $575 + 5750 = \$6{,}612.50$
Year 4	$5750 \cdot 0.05 \cdot 4 = 1150$ $1150 + 5750 = \$6{,}900$

   b. Explain which loan Charlie should choose.
   Charlie should choose the compound interest loan because he will pay back less. The only way he should consider the simple interest loan is if he thinks he will not be able to pay the loan back in 3 years and will need the extra year to pay.

2. The two tables summarize different options Valentina has to invest $1,000. Use the tables to answer the questions.

Simple Interest		
Years Invested	**Calculation**	**Yearly Total (rounded to nearest whole)**
1	$1000 + ($1,000 · 0.07 · 1)	$1,070
2	$1000 + ($1,000 · 0.07 · 2)	$1,140
3	$1000 + ($1,000 · 0.07 · 3)	$1,210
4	$1000 + ($1,000 · 0.07 · 4)	$1,280
5	$1000 + ($1,000 · 0.07 · 5)	$1,350

Compound Interest		
Year	**Calculation**	**Yearly Total (rounded to the nearest whole)**
1	$1,000 (1.07)	$1,070
2	$1,000 (1.07) (1.07)	$1,145
3	$1,000 (1.07) (1.07) (1.07)	$1,225
4	$1,000 (1.07) (1.07) (1.07) (1.07)	$1,311
5	$1,000 (1.07) (1.07) (1.07) (1.07) (1.07)	$1,403

7.8.4 COLLABORATION: COMPARING INTEREST

TEACHER MOVES

The focus of this benchmark is on simple interest. The purpose of this component is not for students to understand how to perform calculations or solve problems with compound interest. Notice those calculations are already done for them. Instead, it is to expose them to the fact that there is another form of interest other than simple interest. They will learn more about compound interest in Algebra 1.

TEACHER MOVES

Think-Pair-Share

Students discuss the similarities and differences in simple and compound interest. Ask students to initially read the problems independently and think about their approach before discussing their thoughts and solving the problems with their partners. These discussions will build student confidence and understanding.

QUESTIONING

Consider inviting a financial professional to join the class (virtually or in-person) to share real-world examples of simple and compound interest. Students will benefit from advice regarding investing and borrowing money.

7.8.4 COLLABORATION: COMPARING INTEREST (CONTINUED)

QUESTIONING

Consider using an example to show that although compound interest usually grows faster than simple interest, short-term simple interest with a high interest rate grows faster than short-term compound interest with a low interest rate.

a. How are the two tables similar?
Both tables are computing interest using a rate of 7% and an initial investment of $1,000.

b. How are they different?
The first option uses the simple interest formula and adds the interest to the initial investment. The second option multiplies by the interest, compounded each year.

c. How are these similarities and differences represented in the calculation column for each scenario?
Both columns show an increase to the $1,000 each year. Option 2 is growing at a faster rate than option 1. This investment is worth more after 5 years.

Find another partner pair and discuss the following.

d. How does the growth in an account that calculates compound interest compare to the growth in an account that calculates simple interest?
Compound interest has more growth in interest than simple interest.

7.8.5 Wrap-Up: Cool Down

1. Your friend is absent and will need to make up the lesson from today. Write a quick note to help them understand what simple interest is, how to calculate it, and how to use it to solve problems.

Answers will vary.
Simple interest is a given rate/percentage of a principal or original amount that is accrued over a time period and added to the principal amount. You find the simple interest by multiplying the principal amount by the interest rate and the length in years of the investment (or loan). Once you have the total simple interest, you add it to the principal amount to find a total either investment or loan.

7.8.5 WRAP-UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

For students who might struggle to explain the concept of simple interest in writing, consider allowing them to use speech-to-text software or an audio recorder (i.e. Flip Grid) or solve and label an example.